

Duru Birinci

856-839-8496 | db662@njit.edu | [linkedin.com/in/duru-birinci](https://www.linkedin.com/in/duru-birinci) | github.com/durubirinci

EDUCATION

New Jersey Institute of Technology

Newark, NJ

Bachelor of Science in Computer Science

Expected Graduation: May 2027

- **Relevant Coursework:** Data Structures and Algorithms, Foundations of Computer Science, Programming Language Concepts, Intensive Programming in Linux, Principles of Operating Systems, Cybersecurity, Networking

EXPERIENCE

Software Engineer Intern

May 2026 – August 2026

Fiserv

Berkeley Heights, NJ

- Built and shipped to production an internal Python log analysis tool for banking workstations, parsing hundreds of files at once, each thousands of lines, and returning results in seconds while cutting triage time for support engineers by over 50%.
- Designed a SQLite-backed analysis layer that flags failure-inducing errors and generates plain-language causes, recommendations, and next steps.
- Engineered a C# / .NET desktop launcher that orchestrates startup, environment setup, and dashboard delivery, giving non-technical users one-click access to the full tool.
- Integrated an OpenAI-powered chatbot for natural-language querying of parsed log data, built to Fiserv's Responsible AI Guidelines.
- Engineered a two-stage PII pipeline using regex detection and HMAC-SHA256 tokenization, ensuring account data is tokenized before it reaches storage or the language model.

Nuclear Information Technology Intern

May 2025 – August 2025

PSEG

Salem, NJ

- Built a SQL-backed reporting system tracking 500+ hardware assets by age, last user, and last known location, giving leadership weekly visibility for replacement planning.
- Digitized hundreds of hard-copy operating procedures into a Microsoft SharePoint library and deployed kiosks across multiple plant sites, modernizing their current process.
- Delivered a 3D scanning solution letting supervisors virtually inspect the plant, cutting physical inspection time by 30% while maintaining safety compliance.
- Established the plant's first ServiceNow-powered support desk, delivering real-time IT support for 200+ employees across departments.

PROJECTS

Energy Analytics CLI | C#, .NET, SQLite

GitHub

- Built a cross-platform C# / .NET command-line tool that ingests and analyzes 23,000+ rows of global energy data across 314 countries (1990–2025).
- Designed a SQLite data layer with Entity Framework Core for efficient storage, indexing, and repeatable querying of large datasets.
- Applied insights from summer IT internship at nuclear power plant to explore data-driven solutions for energy policy decisions.

Credit Card Fraud Detection | Python, Pandas

GitHub

- Built machine learning models to detect fraudulent transactions in dataset of 284,807 credit card transactions with 0.17% fraud rate.
- Achieved 96.3% ROC-AUC score and 81.6% fraud detection rate using Random Forest classifier while maintaining 94% precision to minimize false positives.
- Developed end-to-end ML pipeline including data preprocessing, model training, evaluation, and visualization of results.
- Demonstrated practical application of AI in fintech for solving real-world imbalanced classification problems.

TECHNICAL SKILLS

Programming Languages: Python, Java, Typescript, Javascript, SQL, C#

Frameworks & Web: .NET, React, Next.js, Tailwind CSS, HTML/CSS

Backend & APIs: Node.js, FastAPI, Rest APIs

Databases & Infrastructure : PostgreSQL, Redis